HeartSafe Agent – Execution Plan (Nov 5–13)

Nov 9, 2026

Positioning

HeartSafe is an equity-aware trust and coordination layer on top of an existing remote-monitoring platform, not another AI cardiac monitor. Our anchor is Medly (UHN, Toronto): it already collects daily measurements and alerts the care team, and HeartSafe makes those alerts safer and accountable.

1. The monitoring platform (Medly is the main reference) collects daily weight, blood pressure, heart rate and symptoms, and raises its own normal / caution / critical alerts.
2. HeartSafe adds ECG and imaging signals, checks how reliable every input and model output is, and prepares an explainable review.
3. The clinical team reviews, acknowledges and decides. HeartSafe never makes the clinical decision.
4. HeartSafe tracks every escalation to resolution, documents it, and monitors equity across communities.

Who it is for: adults with heart failure enrolled in remote monitoring, including remote and Indigenous communities. The devices and the agent complement scheduled follow-up visits; they do not replace them.

Medly is not open source (no public code or API found), so we build a platform adapter that reads a generic daily export and run it on a simulated feed modelled on Medly. A real deployment would plug in the platform's own export (Medly, TELUS Home Health Monitoring, Cloud DX), for Medly in partnership with UHN.

The reliability and equity monitoring layer is the strongest differentiator. Open item: a connected-therapeutics link.

Goal and MVP scope

By Nov 13 we demo HeartSafe running on a Medly-style feed for a large simulated cohort, with every escalation tracked, an equity audit, and results against the platform alone.

Must work (MVP)

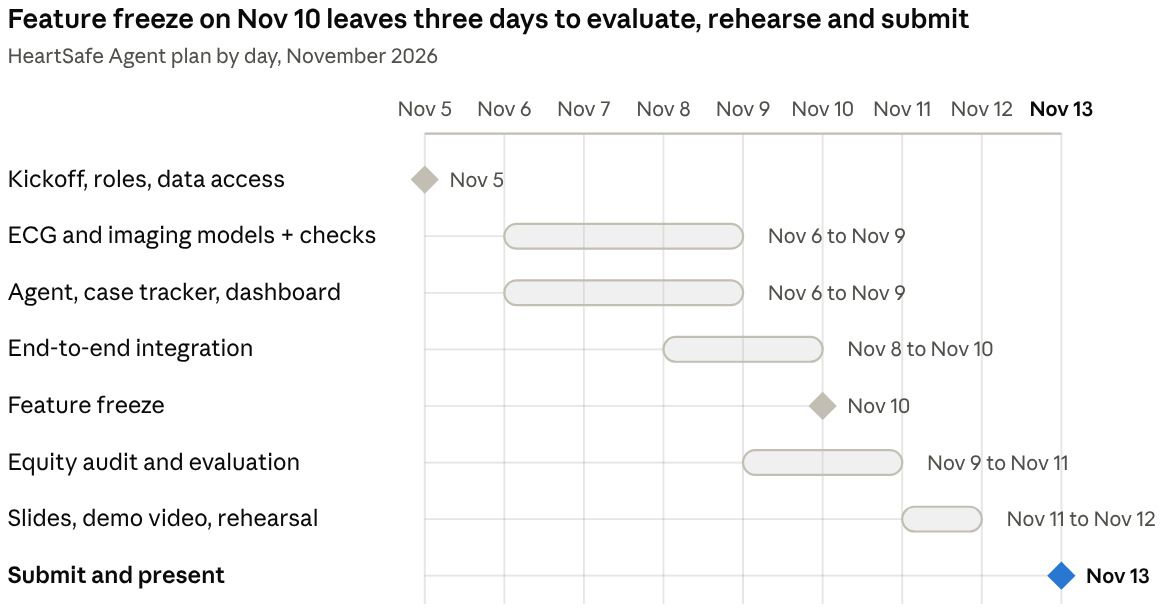
- Platform adapter: reads daily vitals, symptoms and the platform's own alert level. A simulated cohort (2,001 patients × 90 days) follows Medly's published inputs plus the extra inputs cardiologists asked for (diastolic BP, oxygen saturation, more symptoms, diuretic taken, post-discharge labs).
- Inclusion criteria (reviewed with cardiologists): age 18+, a heart-failure admission of 48 h or more, discharged within 14 days, NYHA class II–IV, able to take readings or has a caregiver; excluded: dementia, long-term care, palliative care, transplant list or LVAD, dialysis. 4,400 candidates screened to enrol 2,001. Clinicians can set personal BP and heart-rate limits at enrollment.
- Reliability layer: data-quality checks on platform readings (missing days, implausible values, unit errors, cuff artifacts, new scales verified by a nurse), plus ECG and cardiac MRI model reliability (quality, uncertainty, shift, plausibility), leading to accept / repeat measurement / escalate.
- Explainable review: for each escalation, a structured review (why it was flagged, the trend, reliability caveats, suggested next step). An LLM writes the wording, with a template fallback.
- Follow-up tracker: Created → Acknowledged → Reviewed → Resolved, with deadlines, reminders and an audit log. Only a clinician closes a case.
- Patient loop: after a clinician resolves a case, the instruction goes to the patient; if they do not confirm in time, the caregiver and then the community health worker are contacted. Only the patient side can confirm. Readings are checked 72 h after the action, and the case reopens only if the problem that started it remains or gets worse.
- Equity monitor: missed-day, unreliable-ECG, time-to-acknowledge, acknowledged-in-time and instruction-completed rates by remoteness, sex, age band and device, with bootstrap 95% CIs. Safeguards need administrator approval.
- Evaluation against the platform alone.
- Dashboard: clinician queue, case review, case timeline, patient loop, equity audit.

Large-scale data: PTB-XL (21,799 ECGs) for ECG, M&Ms for cardiac MRI, and the simulated Medly-style cohort for daily vitals.

Stretch: connected-therapeutics ideas (missed readings as an adherence signal), real notifications.

Everyone codes against one shared interface: each model returns `{prediction, probability, uncertainty, quality, shift_score, metadata}`. That lets the agent be built on mock outputs from day one.

Timeline



execution timeline · Nov 5–13

Models and the agent are built in parallel on mock outputs (Nov 6–9) so integration can start Nov 8. Nothing new is added after the Nov 10 freeze; the last three days are evaluation, slides and rehearsal.

Day	Steps
Thu Nov 5	Kickoff; request M&Ms access; set up shared repo; download PTB-XL and PhysioNet 2017; agent skeleton on mock outputs
Fri Nov 6	Start nnU-Net training on the cluster; train the ECG baseline; policy engine and case tracker
Sat Nov 7	Imaging plausibility and shift checks; ECG quality, uncertainty and shift test; LLM tools and review drafts
Sun Nov 8	Integration: swap mocks for real outputs; dashboard views
Mon Nov 9	Volumes/EF, uncertainty maps, subgroup metrics; simulated cohort and fault injection; reminders and audit log

Day	Steps
Tue Nov 10	Safeguard loop; baseline vs agent runs; bug fixes; demo review; feature freeze
Wed Nov 11	Results figures; evaluation table; demo script; record the backup video
Thu Nov 12	Final slides; full rehearsal; fixes only
Fri Nov 13	Submit and present

Build steps

Tick these off as they land; each has a concrete done-check.

Imaging

- [x] ~~Download M&Ms, keep vendor and centre metadata; pick a subset that fits the GPU budget~~
- [x] ~~Train nnU-Net (2d config, one fold, reduced epochs) on the cluster; done = Dice reported per vendor~~
- [x] ~~Compute LV/RV volumes and EF; softmax-entropy uncertainty map per case~~
- [x] ~~Plausibility rules (EF 0–100, volume ranges, one connected LV component) and vendor-shift score~~
- [x] ~~Export cached outputs for every test case in the shared interface format~~

Imaging results (Nov 7, SDRE H200; nnU-Net 2d, fold 0, 250 epochs in 63 min):

Vendor (held out)	Seen in training	n	Dice LV	Dice MYO	Dice RV	LVEF error (points)
Siemens (A)	yes	20	0.936	0.856	0.893	3.6
Philips (B)	yes	50	0.920	0.874	0.894	4.4
GE (C)	no	50	0.890	0.826	0.858	5.3
Canon (D)	no	50	0.904	0.824	0.875	3.8
All		170	0.908	0.843	0.878	4.4 (median 3.0)

- Reliability policy (decided Nov 7): scanner shift alone is a caveat on the review, not an escalation; a severe shift (beyond 99% of clean validation exams from all four vendors, score 14.3) is always escalated. Thresholds come from clean data only.

- Policies compared on 170 clean + 85 fault-injected exams:

Policy	Faults caught	Clean exams flagged	Escalations appropriate	Wrong predictions missed
No agent (baseline)	0%	0%	–	60
Shift always escalates	100%	55%	57%	5
Shift = caveat	77%	21%	75%	12
Shift = caveat, severe shift escalates (chosen)	92%	25%	75%	9

- Chosen policy by fault: noise and slice drop 100%, contrast 90%, blur 76%.
- Fixed on the way: 3 held-out subjects had skewed NIfTI orientation that nnU-Net could not read; converter now cleans it.
- Outputs: 170 clean + 255 fault-injected exam JSONs in the shared format, with uncertainty maps.

ECG

- [x] ~~PTB-XL: one binary task (e.g. normal vs abnormal) on lead I, patient-wise official folds~~
- [x] ~~Small 1D CNN with MC-dropout or a 5-model ensemble for uncertainty~~
- [x] ~~Signal quality index (e.g. NeuroKit2) and heart-rate plausibility~~
- [x] ~~Map PhysioNet 2017 labels to the same task, resample, and measure the performance drop (device shift)~~
- [x] ~~Embedding-distance shift score; export cached outputs~~

ECG results (Nov 7, run on SDRE H200, code in `ecg/`):

- PTB-XL lead I, 21,791 ECGs, folds 1–8 train, 9 calibrate, 10 test; 5-model 1D-ResNet ensemble.
- Test AUROC 0.887 (95% CI 0.874–0.901). PhysioNet 2017 AliveCor: AUROC 0.673 overall, 0.776 for normal vs AF only.
- Reliability v2 (template match, baseline, QRS-band and high-frequency power; calibrated entropy uncertainty; embedding + recording-feature shift score; all thresholds from clean validation ECGs): injected faults 100% caught (v1 72%), 81% of clean ECGs accepted (v1 86%).
- Accuracy on accepted ECGs 0.82 vs 0.79 on all; the most certain half reaches 0.95.

- Device shift score separates AliveCor from PTB-XL with AUROC 0.94; 90% of noisy AliveCor recordings flagged.
- Limits: PTB-XL expert-annotated real noise on lead I is flagged only 23% vs 16% for unannotated ECGs; on AliveCor, uncertainty does not separate right from wrong (AUROC 0.50).
- Cached outputs: one shared-format JSON per ECG (clean, faulty, PhysioNet 2017). Agent's ecg_max_uncertainty (0.35) needs the calibrated value 0.98.

Agent and dashboard

- [x] ~~Policy engine: thresholds in a YAML config, deterministic accept / repeat / escalate~~
- [x] ~~LLM tool-calling orchestrator (quality, uncertainty, shift, plausibility, draft-review tools), first on mock outputs~~
- [x] ~~Case tracker in SQLite with the four states, simulated clock, stalled-case reminders, audit log of every action~~
- [x] ~~Dashboard views: clinician queue, case timeline, equity audit with an "Approve safeguard" button~~
- [x] ~~Template fallback for review drafts if the LLM call fails~~

Agent results (Nov 9 final run, after the reopening fix; simulated cohort of 2,001 patients × 90 days modelled on Medly, real ECG and MRI pipeline outputs, same simulated patients and nurses for both arms, no safeguards):

Measure	HeartSafe	Platform alone
Median hours, first alert to acknowledgement (true episodes)	7.7	13.7
True episodes acknowledged within 24 h	74%	70%
Alerts caused by faulty readings stopped	86% (another 10% held for a nurse scale check)	0%
Nurse queue items per 100 patient-months	86	104
False alarms per 100 patient-months	59	76
Escalations resolved within deadline	97%	86%

Measure	HeartSafe	Platform alone
Median hours to resolution	42.5	44.8
Stalled cases found	487 of 491	0 of 615
Patient side completed the care team's instruction	96%	76%
Remote patients who completed it	94%	65%

- Reopening fix (Nov 9): a case used to reopen when any alert fired at the 72 h check; 63% of those reopenings happened outside a true episode. Now each loop remembers the rules that started its case and reopens only if one still fires or the readings get worse; alerts during the plan that are not worse are added to the case as evidence. Clinicians also set personal BP / heart-rate limits at enrollment for patients whose usual value sits near a default limit (19% of patients, mostly low BP). Reopenings outside a true episode: 9%.
- Real model evidence inside the agent: ECG sensitivity 1.00, specificity 0.84 (faulty ECGs kept out of the evidence; 84% of clean ECGs used). MRI: 48% of unreliable results flagged, 84% of good ones accepted (only 58 unreliable exams reached the agent, so this number is noisy). The agent uses each pipeline's calibrated decision.
- LLM orchestrator: Claude calls read-only tools (case facts, data quality, ECG, MRI, follow-up) and writes the review via submit_review; decisions stay deterministic; each run is in the audit log; template fallback when there is no key. Tested with a stand-in client; the live test (agent/llm_check.py, 5 cases, checks for invented numbers) is ready and waits for an API key.
- Dashboard: patient journey, clinician queue, patient loop, equity audit with approve-and-apply safeguards, results vs the platform alone, model reliability (ECG and imaging results, policy comparison).
- Demo patient (Mary) follows the demo script below: one escalation, one patient loop, no reopenings.
- Assumptions to state on the slide: patient, caregiver and nurse response times, forgetting rates, which patients get faulty ECGs or unfamiliar scanners (agent/config.yaml). The simulation does not model the effect of treatment.

Equity and evaluation

- [x] ~~Subgroup metrics with bootstrap 95% CIs for both models~~

- [x] ~~Gap rule: flag a subgroup when its CI sits below the overall estimate; propose the safeguard~~
- [x] ~~Simulated cohort and baseline runs (see Evaluation)~~

Equity results (Nov 7), bootstrap 95% CIs, gap = subgroup CI entirely worse than overall; every safeguard waits for administrator approval:

- Imaging (LV Dice, LVEF error by vendor, centre, sex, age band): 3 gaps, all on LV Dice: GE 0.890 (never seen in training), centre 4 (the GE centre) and age 60+ 0.888, vs 0.908 overall. No LVEF-error gap.
- ECG (accuracy, unreliable-data rate by sex, age band, device, site): 6 gaps, including lower accuracy for age 40–59 and device CS-12 E, and higher unreliable-data rates for age 60+, device CS-12 and site 2. No sex gap in either model.
- Agent equity monitor (Nov 9): 7 gaps. Remote patients: 40% of ECG checks unreliable, 16.2 h to acknowledgement (urban 10.8 h), 47% acknowledged within the standard deadline (urban 76%), 94.4% completed the instruction (urban 96.6%). Two safeguards proposed for remote patients (tighter acknowledgement deadline; contact the caregiver or community health worker twice as early) plus 4 program recommendations (device, connectivity and ECG coaching outreach).

Safeguard before/after for remote patients (both safeguards approved, same cohort, Nov 9 test run; the live approval is clicked in the dashboard for the demo):

Remote patients	No safeguards	Both safeguards
Mean hours to acknowledgement	16.2 (gap)	10.6 (no gap)
Acknowledged within the standard deadline	47%	91%
Completed the care team's instruction	94.4% (gap)	95.2% (no gap)
Median hours from instruction to action	19.7	17.3
Cost		+389 reminders, +265 helper contacts, +0.3 cases and no extra false alarms per 100 patient-months

After the remote gaps close, rural patients become the new hours-to-acknowledgement gap (12.8 h), so the monitor proposes the next safeguard: the loop keeps running.

Evaluation

We compare the agent with a no-agent baseline (every prediction accepted, no follow-up tracking) on one simulated cohort built from the held-out test sets. To get ground truth for "unreliable", we inject known faults into a share of cases: added noise, flat or clipped ECG segments, cross-device ECGs, and out-of-vendor or degraded MRI slices.

Metric (from abstract)	How we measure it
Detection of unreliable predictions	Sensitivity / specificity of repeat-or-escalate on injected faults
Appropriate escalation	Share of escalations where the model was wrong or the input was faulty
Subgroup gap detection	Gaps flagged vs gaps present in subgroup CIs
Unresolved cases found	Stalled cases flagged vs cases simulated to stall
Follow-up completion	% of escalations resolved within the simulated window
Time to resolution	Median simulated days, agent vs baseline
Extra false alarms	Added escalations after the safeguard is approved
Patient follow-through	Share of resolved cases where the patient side completed the instruction; share of reopenings outside a true episode

Clinician response delays are simulated from a simple assumed distribution; we state that assumption on the slide.

Demo and presentation

The talk follows one patient through the monitoring platform plus HeartSafe, then zooms out to the equity audit and the numbers. HeartSafe is presented as the layer that makes platforms like Medly safer, not a replacement for it.

1. Problem: heart failure care in remote communities; monitoring platforms work, but alerts depend on reliable data and timely follow-up.
2. What platforms like Medly already do (credit them), and the gap: unreliable inputs, alerts without a structured review, lost follow-ups, no subgroup monitoring.
3. Live demo, one patient in a remote community:
    a. Day 20: the platform flags +2.6 kg overnight. HeartSafe sees a jump with no symptoms and a steady heart rate (a new scale?) and asks for a repeat. The jump persists, a nurse calls and confirms the new scale, and the readings are re-baselined; the next weight alerts are filtered.
    b. Days 44–46: no readings; an outreach case goes to the nurse.
    c. Day 50: breathlessness and a good-quality abnormal ECG; critical escalation with an explainable review.
    d. The case stalls, a reminder fires, and the clinician acknowledges and resolves it; every step is in the audit log.
    e. The care team's instruction goes to Mary. She misses the message (remote), her caregiver is contacted, confirms and acts. Alerts on days 53–54 are added to the case while the plan runs instead of opening new cases.
    f. MRI at the regional centre (a scanner the model never saw) has high uncertainty; clinician review requested.
    g. Day 55: readings checked after the action have improved; the loop closes.
4. Equity audit: remote patients wait longer and complete fewer instructions; two safeguards are proposed, an administrator approves them in the dashboard, and the re-run shows the remote gaps closing and what it costs.
5. Results against the platform alone, with the simulation assumptions stated.
6. Responsible AI: the clinician decides, HeartSafe complements follow-up visits, and data use follows OCAP® in partnership with communities.

Record a 2–3 minute backup video of the demo on Nov 11 in case the live run fails.

Risks and cut list

The biggest risks are data access and nnU-Net training time, so both start on Nov 6.

Risk	Fallback
M&Ms download needs registration and may be delayed	Request access today; fall back to a smaller public cardiac MRI set with vendor info
nnU-Net training too slow	2d config, one fold, fewer epochs; demo runs on cached predictions
Integration slips past Nov 10	Demo on cached model outputs; the agent logic stays live
LLM API fails or costs too much	Template-based review drafts, same flow
Subgroup CIs too wide to show a gap	Report it honestly; demonstrate the safeguard loop on the largest observed gap
Nov 11 is Remembrance Day	Confirm everyone's availability at kickoff

If we are behind, cut in this order: real notifications → live LLM root-cause text → live imaging inference → ECG Grad-CAM. Never cut reliability checks, escalation, follow-up tracking or the equity loop; they are the pitch.

Review sessions and submission checklist

Three short review sessions, so feedback lands before each decision point.

- **Nov 6 – scope check (20 min):** Is the MVP the right size? What do judges weigh most? Are the escalation rules clinically sensible?
- **Nov 10 – demo review (30 min):** Walk through the end-to-end flow at feature freeze.
- **Nov 12 – full rehearsal (45 min):** Timed run of slides and demo.

Before submitting on Nov 13

- [ ] Confirm the exact deadline time, time zone and submission format
- [ ] Code repo with README, setup steps and the config file of thresholds
- [ ] Slides (PDF copy too)
- [ ] Backup demo video
- [ ] Evaluation table and subgroup results with CIs
- [ ] Responsible-AI note: de-identified public data only, clinician makes every decision, audit log
- [ ] Dataset and method citations (PTB-XL, PhysioNet 2017, M&Ms, nnU-Net)

- [ ] Team number and contact details on the title slide